DATA ITEM DESCRIPTION

Title: Aircraft Structural Integrity Program (ASIP) Master Plan

Number: DI-SESS-81956A
Approved Date: 20181126
AMSC Number: F9995
Limitation: N/A
DTIC Applicable: No
GIDEP Applicable: No
Preparing Activity: 11 (AFLCMC/EZFS)
Project Number: SESS-2018-050
Applicable Forms: N/A

Use/Relationship: The Aircraft Structural Integrity Program (ASIP) Master Plan is essential to document the tasks and elements of MIL-STD-1530, *Aircraft Structural Integrity Program (ASIP)*, that are planned or accomplished in the development (design, analysis, test, and force management development) and sustainment (force management execution) of the aircraft system. (Copies of this document are available online at https://quicksearch.dla.mil.)

a. This DID contains the format, content, and intended use information for the data deliverable resulting from the work task described in 5.1.1 of MIL-STD-1530.

b. This DID supersedes DI-SESS-81956.

Requirements:

1. Reference documents. The applicable issue of the documents cited herein, including their approval dates and dates of any applicable amendments, notices and revisions, shall be as specified in the solicitation or contract.

2. Format. The contractor's electronic format for the ASIP Master Plan is acceptable.

3. Content. The ASIP Master Plan shall depict the ASIP elements planned or accomplished in the development, certification, and sustainment of the aircraft structure. The required elements of the ASIP are defined in MIL-STD-1530. Discussion shall be provided showing how the ASIP Master Plan was developed, its unique features, exceptions to the requirements of the standard, and any structural problems anticipated or special maintenance actions required to meet the service life. The development of the schedule shall consider all interfaces, the impact of schedule delays (e.g., delays due to test failure), mechanisms for recovery programming, and any other problem areas. Each section of the ASIP Master Plan shall include the information outlined below. Each section of the ASIP Master Plan will also contain an appendix with more detailed information, if necessary.

TABLE OF CONTENTS. This section of the ASIP Master Plan shall include: Table of Contents and Table of Changes.

1.0 INTRODUCTION. This section of the ASIP Master Plan shall contain an overview of the aircraft program; background of the aircraft development program; background on the sustainment strategy; description of the aircraft; various configurations; three-view drawings; pertinent performance information to include design load factors, flight envelopes, weights, etc.; data and other general information. Report detailed information in Appendix A.

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2.0 SUMMARY. The plan shall contain a summary section that briefly presents the important aspects of the particular structural integrity program. This section of the plan will address such items as deviations identified as Not Programmed Task elements (for specific Program Tasks see MIL-STD-1530, Table I) and approved deviations from the general requirements; possible, past, or current problems; schedule slippages; etc. Report detailed information in Appendix B.

3.0 DISCUSSION OF ASIP BY TASK. This section of the plan will discuss the program application of each element contained within the five major ASIP tasks as specified in MIL-STD-1530, Table I. The discussions will be in summary form with supporting detailed information placed in Appendix C. Further details concerning the content of section 3.0 of this plan are contained in the following subparagraphs.

3.1 TASK I, DESIGN INFORMATION. This section of the plan will contain individual subparagraphs for the elements of Design Service Life and Design Usage; Structural Design Criteria; Durability and Damage Tolerance Control; Corrosion Prevention and Control Program (CPCP); Nondestructive Inspection (NDI) Program; and Selection of Materials, Processes, Joining Methods and Structural Concepts (see 5.1 of MIL-STD-1530). The information to be provided in each element shall include the scope and application, deviations, problem areas, status (indicating amount completed) and schedule slippage.

3.2 TASK II, DESIGN ANALYSES AND DEVELOPMENT TESTING. This section of the plan will contain individual subparagraphs for the elements of Material and Structural Allowables, Loads Analysis, Design Loads/Environment Spectra, Stress and Strength Analysis, Durability Analysis, Damage Tolerance Analysis, Corrosion Assessment, Sonic Fatigue Analysis, Vibration Analysis, Aeroelastic and Aeroservoelastic Analysis, Mass Properties Analysis, Survivability Analysis, Design Development Tests, Structural Risk Analysis and Economic Service Life Analysis (see 5.2 of MIL-STD-1530). The information to be provided in each element shall include the scope and application, deviations, analyses and testing problems, status (indicating amount completed) and schedule slippage.

3.3 TASK III, FULL-SCALE TESTING. This section of the plan will contain individual subparagraphs for the elements of Static Tests, First Flight Verification Ground Tests, Flight Tests, Durability Tests, Damage Tolerance Tests, Climatic Tests, Interpretation and Evaluation of Test Findings, and Resolution of Test Findings (see 5.3 of MIL-STD-1530). The information to be provided in each element shall include scope and application, deviations, significant testing problems, test results and evaluation, status (indicating amount completed) and schedule slippage.

3.4 TASK IV, CERTIFICATION AND FORCE MANAGEMENT DEVELOPMENT. This section of the plan will contain individual subparagraphs for the elements of Structural Certification, Strength Summary and Operating Restrictions (SSOR), Force Structural Maintenance Plan (FSMP), Loads/Environment Spectra Survey (L/ESS) System Development, Individual Aircraft Tracking (IAT) System Development, Force Management Database Development, and Technical Orders (TOs) (see 5.4 of MIL-STD-1530). The information to be provided in each element shall include scope and application, deviations, restrictions imposed, actions needed to remove any restrictions, Aircraft Structural Integrity Management Information System (ASIMIS) interface, data acquisition systems, economic/sensitivity studies, status (indicating amount completed) and schedule slippage.

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3.5 TASK V, FORCE MANAGEMENT EXECUTION. This section of the plan will contain individual subparagraphs for the elements of the L/ESS Execution, IAT Execution, Durability and Damage Tolerance Analysis (DADTA) Updates, L/ESS and IAT System Updates, NDI Updates, Structural Risk Analysis Updates, CPCP and Corrosion Assessment Updates, Analytical Condition Inspection (ACI), FSMP Updates, TO Updates, Repairs, Force Management Database Execution, Structural Certification Updates, Economic Service Life Analysis Updates, and Others as required (see 5.5 of MIL-STD-1530). The information to be provided in each element shall include scope and application, deviation, economic analyses, status (indicating amount completed) and schedule slippage. The critical areas and inspection periods to be cited shall be restricted to those identified in Task IV and applicable structural deficiencies identified during service.

4.0 SCHEDULES, RESPONSIBILITIES, AND FLOW DIAGRAMS. The ASIP Master Plan shall contain specific schedules, responsibilities, flow diagrams, and other information needed to define the scope and execution of the ASIP. The plan shall depict the time-phased scheduling and integration of all required ASIP tasks for design, development, certification and force management of the aircraft structure. Report detailed information in Appendix D. This information will include:

a. Checklist of ASIP Elements indicating Programmed, Not Programmed, Active, and Effort Completed (see Table I in MIL-STD-1530 for list of Tasks/Elements).

b. Flow Diagram showing the particular system program status for each Element of each of the five major tasks.

c. ASIP Master Schedule.

d. Delineation of ASIP responsibilities (Air Force Materiel Command (AFMC), Air Force Life Cycle Management Center (AFLCMC), and using command) together with milestones.

e. Detailed schedules and flow diagrams for the actual operational usage program (to include: flight data recorder procurement and installation, data collection, ASIMIS or contractor provided software interface, storage, dissemination, and analysis) for fleet tracking.

f. Modification programs resulting from the findings of any aspect of the ASIP with a discussion of requirements and schedule for the modification.

5.0 FUNDING. The ASIP Master Plan shall include the costs for the ASIP. Subparagraph 5.1 will contain cost data supplied by the contractor. Subparagraph 5.2 will contain organic cost data supplied by the internal Air Force activity. These costs shall include initial estimates together with updated cost information and when possible will be provided separately for each element of ASIP. The costs shall be presented both for actual expenditures and projected requirements as a function of fiscal year for the next five years. Additional cost information must be provided as necessary and available. Report detailed information in Appendix E.

6.0 APPENDICES. The appendices of this plan shall include the supporting data, rationale, decisions, engineering detail, etc., necessary to complement the summary of write-ups presented in sections 1.0 through 5.0 of the ASIP Master Plan. The individual appendices will include the following detailed information as outlined below.

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6.1 Appendix A – Section 1.0 supporting data shall include:

a. Brief history of development.

b. Principal aircraft dimensions.

c. Illustration of major structural components and associated coordinates (numbered wing stations, etc.).

d. Distribution of aircraft by command by Mission Design Series (MDS).

e. Flight hour distribution by MDS. Identify high time aircraft and force average.

6.2 Appendix B - Section 2.0 supporting data shall include:

a. Rationale, discussion of background, impact on the force and copies of official Headquarters (HQ) United States Air Force (USAF) documentation for all approved deviations. For each ASIP element designated "Not Programmed," discuss rationale, background, impact on the force and ASIP objectives and provide copies of any official documentation.

b. Brief discussion of in-service problems, fixes, important inspections and findings.

c. Tabular summary of safety and economic life limits by structural detail. The details to be reported shall be restricted to those identified in Task IV and appropriate service revealed structural deficiencies.

6.3 Appendix C - Shall contain the detailed supporting data for all ASIP elements segregated by Task I through Task V as discussed in Section.3.0.

6.4 Appendix D - Shall contain the detailed supporting data for Section 4.0.

6.5 Appendix E - Shall contain the detailed supporting data for Section 5.0.

6.6 Appendix F - Validation Report. This appendix shall contain a copy of the annual validation report, if applicable, and the coordination sheet for the initial ASIP Master Plan and each subsequent revision of the ASIP Master Plan.

6.7 Appendix G - ASIP Master Plan Distribution. This appendix shall contain the Distribution List of the initial ASIP Master Plan and each subsequent revision of the ASIP Master Plan.

7.0 BIBLIOGRAPHY – Will contain a Master list of all documents referenced in the ASIP Master Plan, listed by section and ASIP task (see Appendix C).

End of DI-SESS-81956A.

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